Software Test Plan

Online Exam Portal | Version 1.0 | 13 November 2026

Online Exam Portal | Team 6 | PES University

Prepared for project documentation and evaluation

Document Control

Field	Value
Version	1.0
Date	13 November 2026
Project	Online Exam Portal — Team 6, PES University
Test catalogue	83 cases in accompanying XLSX
Automated baseline	114 Django tests passed in supplied execution record

1. Introduction

1.1 Purpose

Defines test scope, strategy, resources, environment, entry/exit criteria, security validation and reporting for the Online Exam Portal.

1.2 Scope

Covers accounts, exams, attempts and results: registration, authentication/RBAC, question/exam management, timed attempts, answer saving, submission/evaluation, results and instructor reports.

1.3 References

• Online Exam Portal SRS v1.1 Draft plus Security Objectives Addendum.

• Online Exam Portal SAD plus Implementation Alignment Addendum.

• Online_Exam_Portal_Test_Cases_83.xlsx (83 detailed cases).

• Django source under Source-Code/ and supplied QA reports.

2. Test Items

• accounts: registration, login/logout, roles, sessions.

• exams: question/option CRUD, scheduling, publishing, ownership.

• attempts: start, timer/deadline, answer persistence, submit, scoring, immutability.

• results: student results/history and instructor reports/review.

• Database migrations, Django URLs/views/forms/templates.

3. Features to Be Tested

3.1 Functional

Req	Feature	Verification focus
FR-1	Registration	Valid/invalid student and instructor registration.
FR-2	Authentication/RBAC	Login/logout, generic errors, role restrictions, sessions.
FR-3	Questions	MCQ validation, ownership, edit/delete restrictions.
FR-4	Exams	Create, schedule, publish, validation and ownership.
FR-5	Attempts	Availability, one attempt, instructions, timer, navigation.
FR-6	Submission	Manual/automatic submit, deadline, locking, evaluation.
FR-7	Results	Student history, question review and instructor reports.

3.2 Non-functional

Req	Acceptance target from SRS
NFR-1	Normal pages <3s; result/evaluation <2s for up to 50 questions; 10 concurrent students.
NFR-2	Persist submissions before result display; protect submitted attempts and related exam data.
NFR-3	Salted hashes, server-side RBAC, CSRF, ORM, answer-key confidentiality, valid sessions.
NFR-4	Current Chrome/Firefox, usable at 360px+, exam start within 3 clicks, consistent messages.

4. Features Not to Be Tested

• Payments, external identity providers, email/SMS integrations, proctoring and native mobile apps (not in supplied scope).

• Production infrastructure certification and load beyond the SRS target.

• A separate public REST API; the application is primarily server-rendered Django.

5. Test Approach

5.1 Security Validation

Dedicated validation of NFR-3 and the security objectives: account confidentiality, role/data isolation and exam integrity.

ID	Check	Expected result
SEC-01	Password storage	Django salted hash; no plaintext.
SEC-02	Role authorization	Unauthorized role denied server-side.
SEC-03	Object ownership	No cross-instructor access or modification.
SEC-04	CSRF	State-changing POST without valid token rejected.
SEC-05	Answer confidentiality	Correct answers not exposed during active attempt.
SEC-06	Session	Protected pages require session; logout invalidates; cookie policy respected.
SEC-07	Submitted data	Submitted attempts/results cannot be changed by student/instructor.
SEC-08	Input/database safety	Invalid input rejected; ORM used for user-driven database access.

Security exit rule: any authorization bypass, answer-key exposure, plaintext password storage or unauthorized modification of submitted results is release-blocking.

5.2 Unit Testing

Use Django tests for isolated models, forms, services, views and validation. Execute: python manage.py test accounts exams attempts results. The bare default command was observed to discover zero tests in the supplied environment.

5.3 Integration Testing

Verify interactions across authentication, question/exam persistence, attempt creation, response saving, evaluation and result retrieval.

5.4 System Testing

Run browser workflows using separate Student and Instructor accounts; capture screenshots/logs for each executed scenario.

5.5 Performance Testing

Measure page latency, evaluation time for up to 50 questions and at least 10 concurrent students. Record machine, browser, dataset, concurrency and repetitions.

5.6 Usability and Compatibility

Test current Chrome/Firefox, 360px and desktop widths, three-click exam start and consistent validation messages.

5.7 Regression Testing

After fixes, rerun affected cases and full automated suite; record commit/build and outcomes.

5.8 Test Data

• Synthetic student/instructor accounts and test emails.

• Open/closed/upcoming exams; boundary duration and schedule values.

• Valid/invalid MCQ options, marks and answer selections.

• No real credentials or personal data in committed fixtures.

5.9 Execution and Reporting

The 83-case XLSX is the detailed catalogue. Record Actual Result, Pass/Fail, tester, date and evidence for each execution. Blank result cells in the supplied workbook must not be inferred as passed.

5.10 Defect Management

Log ID, severity, priority, affected requirement/case, steps, expected/actual result, evidence, owner, status and retest. Critical security/data-integrity defects block release.

5.11 Entry Criteria

• SRS/SAD baseline reviewed; dependencies installed and migrations applied.

• Test accounts/data prepared; build/commit recorded.

• No unresolved merge conflicts; test cases reviewed.

5.12 Exit Criteria

• Critical/high cases executed or explicitly marked blocked/not run.

• No open critical security/data-integrity defects.

• Automated results and failures documented.

• Performance evidence includes measured values.

• Coverage reports identify source scope and commit.

• System evidence, deviations and known limitations reviewed.

5.13 Suspension / Resumption

Suspend for startup failure, unstable test database, corrupted fixtures or critical blocker. Resume after fix, migration/smoke checks and test-data reset.

5.14 Roles

Role	Responsibility
Project Manager	Schedule, scope, issue tracking and release coordination.
Developers	Fix defects, maintain tests and provide technical evidence.
QA/Documentation Lead	Maintain cases, coordinate execution and report results.
All members	Review requirements and participate in system/viva evidence.

5.15 Risks

• SQLite local results may not represent production concurrency.

• Clock-dependent tests need controlled timestamps.

• Unexecuted manual tests remain Not Run/Pending.

• Performance claims require recorded measurements.

6. Test Deliverables

• Approved test plan and 83-case workbook.

• Unit/integration/system/security/performance/regression evidence.

• Unit, system validation, line coverage and branch-enabled coverage reports.

• Defect log, retest evidence and final test summary.

7. Test Environment

Component	Environment
Application	Django; accounts, exams, attempts, results.
Runtime	Python 3.11+ per SRS; record exact runtime per run.
Database	SQLite via Django ORM; Django test database for automated tests.
Client	Current Chrome and Firefox; record OS/device.
Tools	Django test framework, Coverage.py, browser tools, GitHub Actions.

8. Schedule and Milestones

The uploaded snapshot does not contain an approved test calendar. The team should assign dates for plan review, unit/integration execution, security validation, system/compatibility testing, performance testing and final regression.

9. Requirement Traceability

SRS Appendix C remains authoritative. The table below reproduces the workbook Summary coverage counts.

Req	Requirement	Priority	Cases
FR-1	Registration	High	11
FR-2	Login/RBAC	High	10
FR-3	Question management	High	11
FR-4	Exam management	High	10
FR-5	Exam attempt	High	10
FR-6	Submit/evaluation	High	8
FR-7	Results	Medium	8
NFR-1	Performance	Medium	3
NFR-2	Data protection	Medium	3
NFR-3	Security	High	5
NFR-4	Usability/compatibility	Medium	4

10. Approval

Role	Signature	Date
Project Manager		
QA / Documentation Lead		
Faculty / Reviewer		

Appendix A — Detailed Test Cases

See the accompanying Online_Exam_Portal_Test_Cases_83.xlsx for all 83 detailed cases. Actual Result and Test Result fields are blank in the supplied snapshot and must be populated after execution.

Appendix B — Execution Baseline

A separate supplied execution record showed 114 Django tests discovered and passed with zero failures/errors. Coverage.py reports supplied separately showed 90% line coverage and 88% total coverage in the branch-enabled report. These automated results do not establish that all 83 manual workbook cases were executed.